

Customer Churn Analysis

Tools Used: Excel, SQL, Python and Power BI

Project Type: End-to-End Data Analytics Project

1. Project Overview

Customer churn occurs when customers stop using a company's products or services. This project analyzes churn patterns using customer information such as tenure, contract type, payment method, customer segment, monthly charges and total charges. The project follows an end-to-end workflow using Excel, SQL, Python and Power BI.

2. Project Objective

The main objective is to analyze customer churn patterns and identify customer groups associated with higher churn.

- What is the overall customer churn rate?
- Which contract type has the highest churn?
- Does churn vary by payment method?
- Which customer segments show higher churn?
- How does customer tenure affect churn?
- Do churned customers have different monthly charges?
- Which customer groups can be considered higher risk?

3. Tools and Technologies Used

Tool	Purpose
Microsoft Excel	Data cleaning and initial data analysis
SQL	Data querying and business analysis
Python	Data validation and exploratory analysis
Pandas	Data manipulation and analysis
NumPy	Numerical operations
Matplotlib	Data visualization
Power BI	Interactive dashboard creation

4. Dataset

The dataset contains approximately 1,000 customer records and includes demographic, service, payment and churn information.

Column	Description
Customer_ID	Unique identification number for each customer
Gender	Gender of the customer

Column	Description
Age	Age of the customer
Tenure_Months	Number of months with the company
Contract_Type	Type of customer contract
Payment_Method	Method used for payment
Customer_Segment	Customer category or segment
Monthly_Charges	Monthly amount charged
Total_Charges	Total charges associated with the customer
Churn	Indicates whether the customer churned
Tenure_Group	Group created based on customer tenure
Churn_Flag	Numerical representation of churn status

5. Project Workflow

Raw Dataset → Data Cleaning → Excel Analysis → SQL Analysis → Python Analysis → Power BI Dashboard → Business Insights → Recommendations

6. Data Cleaning and Preparation Using Excel

- Checked the dataset structure.
- Identified and removed duplicate records.
- Checked for missing values.
- Reviewed and formatted data types.
- Standardized column names where required.
- Validated numerical columns.
- Created additional calculated columns for analysis.

Derived Columns

Tenure_Group: Customers were categorized into tenure groups such as 0-12 Months, 13-24 Months, 25-48 Months and 49+ Months.

Churn_Flag: Churn was converted into numerical format: Yes = 1 and No = 0.

7. Excel Analysis

- Total customers
- Churned customers
- Retained customers
- Overall churn rate
- Churn by contract type
- Churn by payment method
- Churn by customer segment

- Churn by tenure group
- Monthly charge comparison
- Revenue-related analysis

8. SQL Analysis

The cleaned dataset was imported into a SQL database. Important queries were organized in Customer_Churn_Analysis.sql.

- Total Customers
- Churned Customers
- Retained Customers
- Overall Churn Rate
- Churn by Contract Type
- Churn by Payment Method
- Churn by Customer Segment
- Churn by Tenure
- Average Monthly Charges
- Revenue Analysis
- High-Risk Customer Combinations

9. Python Analysis

Python was used to validate the cleaned dataset and perform programmatic exploratory analysis using Customer_Churn_Analysis.ipynb.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

- Loaded the cleaned dataset.
- Examined records and dataset shape.
- Reviewed column names and data types.
- Checked missing values.
- Checked duplicate records.
- Performed churn analysis.
- Created visualizations.
- Validated results from Excel and SQL.

10. Power BI Dashboard

An interactive Power BI dashboard was created to communicate key churn findings.

KPI Cards

- Total Customers
- Churned Customers
- Retained Customers
- Churn Rate
- Average Monthly Charges
- Total Revenue

DAX Measures

Total Customers =
DISTINCTCOUNT(Customer_Churn[Customer_ID])

Churned Customers =
CALCULATE([Total Customers], Customer_Churn[Churn] = "Yes")

Retained Customers =
CALCULATE([Total Customers], Customer_Churn[Churn] = "No")

Churn Rate =
DIVIDE([Churned Customers], [Total Customers], 0)

Average Monthly Charges =
AVERAGE(Customer_Churn[Monthly_Charges])

Total Revenue =
SUM(Customer_Churn[Total_Charges])

Dashboard Visualizations

- Churn Rate by Contract Type
- Churn Rate by Payment Method
- Churn Rate by Customer Segment
- Churn Rate by Tenure Group
- Average Monthly Charges by Churn Status
- Interactive filters and slicers

Slicers

- Contract Type
- Payment Method
- Customer Segment
- Gender
- Tenure Group

11. Key Business Insights

Final insights should be based on the actual results obtained from the dataset and dashboard. The analysis identifies patterns and associations, which should not automatically be interpreted as direct causes of customer churn.

12. Business Recommendations

Improve Early Customer Engagement

- Improve onboarding.
- Provide early customer support.
- Offer loyalty benefits during the initial customer lifecycle.

Encourage Long-Term Contracts

- Provide incentives for longer contracts.
- Introduce loyalty benefits and retention offers.

Investigate High-Risk Payment Methods

- Investigate potential payment friction.
- Improve payment convenience.
- Encourage automatic payment options.

Target High-Risk Customer Segments

Customer groups with higher churn rates can be targeted with personalized retention campaigns and offers.

13. Project Limitation

This is a descriptive data analysis project. It identifies patterns associated with customer churn but does not build a machine learning model to predict future churn. The dataset also does not include customer satisfaction, service quality, complaints, competitor activity or detailed support history.

14. Project Structure

```
Customer-Churn-Analysis
├──
│   ├── Raw Data
│   ├── Cleaned Data
│   ├── Excel
│   ├── Customer_Churn_Analysis.xlsx
│   ├── SQL
│   ├── Customer_Churn_Analysis.sql
│   ├── Python
│   ├── Customer_Churn_Analysis.ipynb
│   ├── Power BI
│   ├── Customer_Churn_Analysis.pbix
│   └── README.md
```

15. Skills Demonstrated

- Data Cleaning

- Data Validation
- Exploratory Data Analysis
- Excel Formulas and PivotTables
- SQL Queries and Aggregations
- GROUP BY and CASE Statements
- Python Data Analysis
- Pandas
- Data Visualization
- DAX
- Power BI Dashboard Development
- Business Insight Generation

16. Conclusion

This project demonstrates a complete end-to-end data analytics workflow using Excel, SQL, Python and Power BI. The analysis helps identify churn patterns across customer tenure, contracts, payment methods, customer segments and charges. The final Power BI dashboard provides an interactive way to explore these patterns and support retention decisions.